A-level

ECONOMICS

Paper 1 Markets and market failure

Predicted Paper Set C — Advanced

Diagram and graph focus

MARK SCHEME and MODEL ANSWERS

Total paper mark: 80 | Section A: 40 marks | Section B: 40 marks

General marking guidance

- Examiners reward responses for what candidates have demonstrated, not penalised for omissions.
- Where a response falls between two levels, the mark awarded should reflect the quality of analysis and evaluation, not just the number of points made.
- Diagrams must be accurate and fully labelled to score full marks. A clear verbal description of the diagram may earn partial credit.
- Indicative content is not exhaustive. Alternative relevant arguments and examples should be credited on merit.
- For evaluation questions, credit the quality of judgement — a clear, reasoned conclusion supported by analysis is essential for the top level.

Assessment Objectives (AOs)

AO1 Knowledge and understanding of economic terms, concepts, theories and models.

AO2 Application of economic knowledge and understanding to various contexts.

AO3 Analysis of economic issues, showing understanding of their impact on economic agents.

AO4 Evaluation of economic arguments and use of qualitative and quantitative evidence to support informed judgements.

Section A — Data Response

The UK Energy Price Cap and the Household Energy Market

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0 1	*Calculate the change in the proportion of households in fuel poverty, 2019–2024.*	**2 marks**

Indicative content

The question asks for the change in the proportion itself. Accept EITHER interpretation:

Preferred interpretation — percentage-point change: 14.5% − 10.3% = **+4.2 percentage points** (an increase).

Alternative — percentage change in the proportion: ((14.5 − 10.3) / 10.3) × 100 = (4.2 / 10.3) × 100 = 40.78% ≈ **+40.8%** (an increase).

Mark allocation

- **1 mark** — correct numerical answer (4.2 percentage points or 40.8%; accept 40.7%–40.8%).
- **1 mark** — correctly identifying the change as an increase.

Award 2 marks for either a fully correct percentage-point change (+4.2 pp, increase) or a fully correct percentage change (+40.8%, increase). Do not penalise candidates who use either convention provided the interpretation is clear and the arithmetic is correct. Award 1 mark for a correct figure without the direction stated, or for correct working producing a minor arithmetic error.

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0 2	*Diagram and explanation: how a maximum price below equilibrium creates excess demand.*	**4 marks**

Indicative diagram

A standard supply-and-demand diagram with:

- Axes: Price on the vertical axis, Quantity on the horizontal axis.
- Downward-sloping demand curve (D) and upward-sloping supply curve (S).
- Free-market equilibrium at price P_1 and quantity Q_1.
- A horizontal maximum-price line P_{max} drawn below P_1.
- At P_{max}, quantity demanded Q́ exceeds quantity supplied Q̀.
- Horizontal distance (Q́ − Q̀) clearly marked as excess demand (shortage).

Indicative written explanation

A maximum (or ceiling) price is a legal upper limit on the price at which a good may be sold. If it is set below the free-market equilibrium P_1, firms will be willing to supply only Q̀ at the capped price (moving along the supply curve), while consumers will wish to buy Q́ (moving along the demand curve). Because $\acute{Q} > \grave{Q}$, the market cannot clear at P_{max}: there is excess demand equal to $\acute{Q} - \grave{Q}$. Non-price rationing methods — queues, waiting lists, or in the energy case taxpayer-funded subsidies to suppliers — must emerge to manage the shortage. Extract C confirms this effect in the UK: at the capped price, quantity demanded exceeded what suppliers were willing to supply at unsubsidised cost, requiring £40bn of taxpayer support.

Mark allocation (4 marks)

- **1 mark** — correctly labelled axes (Price and Quantity) with S and D curves and free-market equilibrium (P_1, Q_1).
- **1 mark** — maximum-price line P_{max} correctly drawn below P_1.
- **1 mark** — excess demand ($\acute{Q} - \grave{Q}$) clearly identified on the diagram.
- **1 mark** — written explanation of the mechanism (cap below equilibrium → Qd > Qs → shortage).

A clear verbal description of the diagram may earn up to 3 marks where a visual diagram is absent but the mechanism is fully explained.

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0 3	*Analyse two reasons why the energy price cap may correct or worsen market failure.*	**9 marks**

Reason 1: The cap may CORRECT market failure — equity and protection of vulnerable households

Essential goods such as domestic energy have low price elasticity of demand in the short run — households cannot easily substitute away from heating. Without intervention, a price shock such as the 2022 wholesale gas spike (Extract B: 40 p/therm in 2019 rising to 200 p/therm in 2022) passes almost entirely through to bills, creating a welfare failure in which poorer households face disproportionate hardship. Extract C shows that the lowest decile spent 11% of disposable income on energy in 2022 versus 4% for the top decile — a highly regressive outcome. A price cap redistributes the cost of the shock from households to the government and therefore to the general taxpayer, protecting vulnerable consumers. Fuel poverty rose from 10.3% to 19.8% at the crisis peak but would have been materially higher without the Energy Price Guarantee.

Reason 2: The cap may WORSEN market failure — distorted signals and excess demand

Prices in a competitive market perform a signalling function: a wholesale price spike signals scarcity and incentivises both reduced consumption and investment in energy efficiency and alternative supply. A cap mutes this signal. Extract A notes critics' concern that the cap 'dampens price signals to invest in efficiency'. Extract C confirms the classic textbook consequence: at the capped price, quantity demanded exceeded quantity supplied at unsubsidised cost, generating an effective shortage. The £40bn of taxpayer support required to clear the market represents a large opportunity cost — funds that cannot be spent on healthcare, education or public investment in renewables. By smoothing the short-run pain, the cap may actually delay the long-run structural adjustment the economy needs to make.

Chains of reasoning expected

(Correcting) Cap → lower household bills → higher real disposable income for low-income groups → reduced fuel poverty → improved equity outcome.

(Worsening) Cap → muted price signal → weaker incentive for efficiency investment and demand response → persistence of energy-intensive behaviour → higher long-run emissions and fiscal cost.

Application using the extracts

- **Extract A:** identifies both sides explicitly — the cap 'dampens price signals' yet 'shielded vulnerable households at a critical moment.'
- **Extract B:** wholesale gas rose from 40 to 200 p/therm in 2022 while capped bills stabilised; fuel poverty peaked at 19.8% in 2022.
- **Extract C:** quantifies the mechanism — £40bn taxpayer support, regressive incidence (11% vs 4% of disposable income).

Mark scheme — levels

Level	Marks	Descriptor
Level 3	7–9	Two distinct, fully-developed chains of reasoning — one on each side OR two on the same side — with accurate application to at least two extracts and explicit reference to market failure (equity, information, allocative efficiency, externalities). Sustained economic analysis throughout.
Level 2	4–6	Two reasons identified but at least one is underdeveloped, or only one is explored in depth. Some extract application. Minor inaccuracies.
Level 1	1–3	One reason identified; limited development. Little or no application of the extracts. Market-failure framework weak or absent.

Maximum 6 marks if only one reason is explored in depth. Credit any economically valid second reason — e.g. the cap reducing asymmetric information costs by simplifying consumer choice, or worsening allocative inefficiency by sustaining demand for gas during the transition to renewables.

0 4	*Evaluate the view that maximum prices are the most effective way to protect consumers in essential markets such as energy.*	**25 marks**

Arguments supporting maximum prices (FOR)

- **Direct, immediate consumer protection:** a cap produces an instant reduction in prices paid, bypassing transmission lags. Extract B shows capped bills of £2,500 in 2022 against a counterfactual where the wholesale gas rise from 40 to 200 p/therm would otherwise have driven them far higher.
- **Progressive incidence when essentials are regressive:** Extract C shows the bottom decile spent 11% of disposable income on energy versus 4% in the top decile. A cap disproportionately benefits low-income households.
- **Macroeconomic stabilisation:** reduces headline inflation, supporting real wages, maintaining aggregate demand and reducing the need for rapid monetary tightening.
- **Political and administrative feasibility:** a cap is transparent, easily understood by voters, and quick to implement — unlike means-tested transfers which have high administrative costs and take-up gaps.
- **Inelastic short-run demand:** households cannot quickly switch heating source or insulate homes. Without intervention they would bear the full burden of a supply shock they cannot avoid.

Arguments against / evaluation (AGAINST)

- **Fiscal cost:** Extract C reports £40bn of taxpayer support was required — a large opportunity cost borne by all taxpayers, including those who do not need subsidy.
- **Distorted price signals:** as noted in Extracts A and C, the cap reduces incentives to reduce consumption, invest in efficiency or switch to renewables — slowing the decarbonisation response.
- **Excess demand and rationing:** a cap below free-market equilibrium creates a shortage requiring non-price rationing. In energy, this was handled through taxpayer subsidy to suppliers rather than queues.
- **Poorly targeted:** the cap is per unit, not per household, so wealthy households with larger homes receive the biggest absolute subsidy. A means-tested income transfer would be more efficient in equity terms.
- **Supplier viability:** energy retailers failed in large numbers in 2021–22 when the cap was below costs, creating instability in the market and further fiscal costs through the Supplier of Last Resort process.

Alternative or complementary policies (comparative evaluation)

- Means-tested income transfers (Cost of Living Payments, Warm Home Discount) — better targeting but high administrative costs and imperfect take-up.
- Per-unit subsidies to producers — similar effect without the excess demand, but even larger fiscal cost.
- Investment in insulation and efficiency (ECO scheme, Great British Insulation Scheme) — addresses the structural problem but slow.
- VAT reductions on energy — broad but regressive.
- Regulation: Ofgem standing charge and price cap methodology reforms.

- Long-run supply-side: accelerated renewables, nuclear and storage to reduce gas dependence and wholesale-price volatility.

Expected judgement and conclusion

A top-band answer should reach a reasoned judgement that ‘most effective’ overstates the case. Maximum prices are effective at rapid, universal consumer protection during a crisis, but they are fiscally expensive, poorly targeted and distort price signals. The most effective consumer-protection strategy in essential markets is therefore a layered package: short-run caps to prevent immediate hardship, combined with means-tested transfers for targeted support, and longer-run supply-side policy (efficiency retrofits, renewables) to address the structural cause. The UK experience of 2022–24 is consistent with this layered view — the cap was effective as an emergency tool but is being withdrawn once wholesale gas prices normalise (Extract B: 200 p/therm in 2022 falling to 68 in 2024), suggesting policymakers themselves treat caps as a short-term rather than permanent instrument.

Mark scheme — levels

Level	Marks	Descriptor
Level 5	21–25	Sophisticated analysis and evaluation. Sustained chains of reasoning; accurate diagram(s); substantial use of all three extracts and own knowledge; clear, well-supported judgement that directly addresses ‘most effective’ by comparing caps with at least two alternative policies.
Level 4	16–20	Good analysis and evaluation. Diagram present and mostly accurate. Two or more developed evaluative points on each side. At least one alternative policy considered. Supported judgement, possibly less nuanced.
Level 3	11–15	Sound analysis with some evaluation. Use of extracts is present but may be descriptive. Judgement may be asserted rather than reasoned.
Level 2	6–10	Some relevant analysis. Limited evaluation. Weak or partial use of extracts; comparison with alternatives absent or superficial.
Level 1	1–5	Limited analysis; evaluation absent or generic. Little or no use of extracts.

Section B — Essay

Subsidies, price controls and government failure

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0 5	*Explain, using a diagram, how a per-unit subsidy to renewable electricity producers affects equilibrium price and quantity.*	**15 marks**

Required definitions

- **Per-unit subsidy:** a government payment made to producers per unit of output sold. Its purpose is to reduce the producers' effective marginal cost of production and therefore raise equilibrium quantity supplied.
- **Equilibrium:** the price and quantity at which Qd = Qs and the market clears.

Required diagram

A standard supply-and-demand diagram for renewable electricity showing:

- Axes: Price (£ per MWh) on the vertical, Quantity (MWh) on the horizontal.
- Downward-sloping demand curve D (unchanged by the subsidy).
- Original supply curve S_1 and post-subsidy supply curve S_2 lying vertically below S_1 by the amount of the subsidy per unit.
- Original equilibrium at P_1, Q_1 and new equilibrium at P_2, Q_2 where $P_2 < P_1$ and $Q_2 > Q_1$.
- Producer-received price $P_p = P_2$ + subsidy (above P_1).
- Total cost of the subsidy shaded as the rectangle (subsidy × Q_2) between S_1 and S_2 up to Q_2.
- Incidence of the subsidy between producers and consumers depends on relative elasticities of demand and supply.

Required written explanation

A subsidy reduces producers' effective marginal cost so the supply curve shifts vertically downwards by the subsidy. At any given price producers are now willing to supply a larger quantity. Demand is unchanged, so the market clears at a lower consumer price P_2 and a higher quantity Q_2 — more renewable electricity is produced and it is cheaper to buy. Producers receive P_2 from consumers plus the subsidy from government, so the effective producer price P_p exceeds P_1. The distribution of the subsidy between consumers ($P_1 - P_2$) and producers ($P_p - P_1$) depends on elasticities: an inelastic demand curve transfers most of the benefit to consumers, while an inelastic supply curve retains most of the benefit with producers. Examples: the UK Contracts for Difference scheme for offshore wind; Feed-in Tariffs (2010–2019); the Renewables Obligation.

Welfare and market-failure context

Because fossil generation has a negative externality (CO_2 emissions), the unsubsidised market underproduces renewables relative to the socially optimal level. The subsidy addresses this externality from the positive side: it reduces the relative cost of the cleaner technology, moving production towards the social optimum. A welfare-analysis extension can show the triangle of welfare gain against the rectangle of subsidy cost — net welfare is improved if and only if the external-cost correction exceeds the deadweight loss of the tax needed to fund the subsidy.

Expected content for top-band responses

- Accurate diagram with both S_1 and S_2, P_1/Q_1 and P_2/Q_2, and the subsidy rectangle clearly labelled.
- Clear distinction between consumer price P_2 and producer price P_p.
- Reference to elasticity determining incidence.
- Concrete UK example (Contracts for Difference, Feed-in Tariffs, Renewables Obligation, Boiler Upgrade Scheme for heat pumps).

Mark scheme — levels

Level	Marks	Descriptor
Level 4	13–15	Fully developed explanation with accurate, well-labelled diagram (including producer price, consumer price and subsidy rectangle). Clear chain of reasoning. Reference to elasticity and to a real-world UK subsidy.
Level 3	9–12	Sound explanation with mostly accurate diagram. Minor omissions (e.g. missing subsidy rectangle or producer price). Some real-world application.
Level 2	5–8	Some understanding. Diagram present but shift direction confused, or cost/price distinction muddled. Little application.
Level 1	1–4	Descriptive only. Diagram missing, incorrect shift, or major labelling errors.

0 6	*Evaluate: "Government failure is often worse than the market failure it tries to correct."*	**25 marks**

Key definitions

- **Market failure:** the free market fails to allocate resources efficiently — e.g. externalities, public goods, merit/demerit goods, asymmetric information, monopoly power, inequity.
- **Government failure:** an intervention produces a net welfare loss by creating new distortions larger than the market failure being corrected — e.g. unintended consequences, administrative cost, regulatory capture, information deficits, political short-termism.

Arguments in favour of the statement (FOR — government failure often worse)

- **Unintended consequences of price controls:** the 2022–24 energy price cap cost £40bn of taxpayer money (Extract C) and blunted incentives to conserve or invest in efficiency. It also produced supplier failures in 2021–22, destabilising the market.
- **Subsidy inefficiency and fiscal cost:** early Feed-in Tariffs for solar PV (2010–11) overpaid generators — an administered price well above the falling cost of solar panels locked consumer bills into high payments for 20 years. The scheme was closed in 2019 but the liabilities continue.
- **Information deficits:** regulators cannot easily know the true marginal abatement or production costs that the market would have revealed through prices.
- **Regulatory capture:** producers lobby for subsidies and against their removal — e.g. the persistence of fuel duty freezes since 2011 despite climate objectives.
- **Administrative burden and compliance costs:** Making Tax Digital, the UK ETS auction system, and the complexity of the Warm Home Discount eligibility have all been criticised for high admin overhead relative to benefit.
- **Moral hazard:** bank bailouts after 2008 arguably encouraged further risk-taking; the energy cap may defer essential household energy transitions.

Arguments against the statement (AGAINST — government failure usually less severe)

- **Clear successes of intervention:** the UK plastic bag charge cut single-use bag consumption by over 95%; the Soft Drinks Industry Levy drove significant reformulation; Contracts for Difference auctions have lowered offshore wind strike prices to below wholesale — these are interventions that correct rather than worsen market failure.
- **Scale of market failure can be catastrophic:** climate change is arguably the largest market failure in history (Stern, 2007). Even an imperfect intervention is vastly preferable to inaction.
- **Equity dimension:** free markets are silent on fairness. Extract C shows bottom-decile households spent 11% of disposable income on energy in 2022 — no non-intervention strategy addresses this.
- **Public goods and externalities:** national defence, flood defences, basic scientific research — all under-provided without state finance. Any administrative cost is likely to be dwarfed by the gain from actually providing these goods.

- **Regulatory successes:** ULEZ expansion (2023) reduced nitrogen dioxide concentrations by over 20% in outer London; Clean Air Act predecessors effectively ended London smog.

Analytical framework for judgement

The comparison depends on three variables: (i) the severity of the underlying market failure, (ii) the quality of the policy design, and (iii) the institutional capacity of government. Where market failure is severe (climate change, pandemic response) and design is well-evidenced (UK ETS, plastic bag charge, SDIL), intervention clearly dominates. Where market failure is moderate and policy is poorly designed or politically captured (some historic FiT schemes, blanket energy price caps lasting beyond the acute crisis), government failure can indeed be larger than the problem it addresses.

Expected judgement and conclusion

The best answers reject the word ‘often’ as too strong. Sometimes government failure is worse — badly designed subsidies or price controls sustained beyond their justified window are genuine examples. But in many of the most important UK policy areas — environmental regulation, public health, public goods — the evidence points the other way: intervention has corrected significant market failures at lower welfare cost than inaction. The preferred conclusion is therefore conditional: government failure is possible, sometimes serious, but it is not generally more severe than market failure. Good policy design — sunset clauses, regular review, market-based instruments where appropriate, independent regulators — substantially reduces the risk.

Mark scheme — levels

Level	Marks	Descriptor
Level 5	21–25	Sophisticated, balanced analysis and evaluation. Uses multiple UK examples of both successful and unsuccessful interventions. Clear, nuanced judgement that directly addresses ‘often’. Analytical framework for comparison. Diagrams where helpful.
Level 4	16–20	Good analysis and evaluation. At least two UK examples of price controls or subsidies on each side. Supported judgement, possibly less developed framework.
Level 3	11–15	Sound analysis with some evaluation. One or two UK examples. Judgement may be asserted rather than reasoned.
Level 2	6–10	Limited analysis. Evaluation weak or superficial. Few or no examples.
Level 1	1–5	Descriptive only. Little or no evaluation.

Summary of Marks

Question	Topic	AO focus	Marks
0 1	Quantitative skill — proportion change	AO2	2
0 2	Maximum price — supply and demand diagram	AO1, AO2	4
0 3	Energy price cap — correcting / worsening market failure	AO1, AO2, AO3	9
0 4	Maximum prices as consumer protection — evaluation	AO1–AO4	25
0 5	Per-unit subsidy to renewable producers	AO1, AO3	15
0 6	Government failure vs market failure — evaluation	AO1–AO4	25
TOTAL	**Paper 1: Markets and Market Failure**	**—**	**80**

END OF MARK SCHEME